Question				Marking details	Marks Available AO1	AO2	AO3	Total	Maths	Prac
2 HT	(a)	(i)		$\frac{3.2}{2}$ **or** $\left(\frac{3.2}{8}\right) \times 4$ **or** 2.8 - 1.2 (1) X = 1.6 [cm] (1)		2		2	2	
		(ii)		Substitution into: speed $= \frac{\text{distance}}{\text{time}} = \frac{40}{5}$ (1) = 8 [cm/s] (1) N.B. Substitution into **wave equation** i.e. $f = \frac{v}{\lambda} = \frac{40}{5} = 8$ don't award the first 2 marks Manipulation **and** substitution of $v = f\lambda$ i.e. $f = \frac{8(\text{ecf})}{1.6(\text{ecf from part (i)})}$ (1) = 5 [Hz] (1) If no workings shown answer of 8 on answer line award no marks. If intermediate calculation shown with 8 on the answer line award the first 2 marks. **Alternative:** $= \frac{40}{1.6(\text{ecf})}$ (1) = 25 waves (1) [in 5 seconds] Frequency $= \frac{25}{5(\text{ecf})}$ (1) = 5 [Hz] (1)	1	1 1 1		4	4	

Question				Marking details	Marks Available					
					AO1	**AO2**	**AO3**	**Total**	**Maths**	**Prac**
	(b)			Angle of reflection correct by eye at least 3 wavefronts (1) N.B. point of reflection must coincide with the point of incidence by eye Parallel wavefronts and constant wavelength same as the incident wavelength for all gaps by eye (1) N.B. 2 marks can only be awarded if lines have been drawn with a ruler **Don't award** any marks for refraction		2		2		
				Question 2 total	**1**	**7**	**0**	**8**	**6**	**0**